

RegTrace-Agent: A Closed-Loop Framework for Evidence-Grounded Regulatory Response Review

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Abstract

Financial LLM systems are increasingly asked to decide whether an institution’s response actually resolves a prior review request, yet most financial benchmarks still evaluate static QA, extraction, or classification. We introduce RegTrace-Agent, a framework for evidence-grounded response resolution. Given a reviewer request, an organizational response, and revised artifacts, RegTrace-Agent retrieves evidence, decomposes the request into obligations, verifies visible gaps, constructs scalar/category/natural-language feedback, and optimizes a guarded reviewer policy. We instantiate the framework on SEC comment-letter review and construct RegTrace-SEC, 472 public EDGAR examples with amended-filing evidence, visible-evidence labels, feedback fields, and later same-topic SEC follow-up signals. Experiments show that amended evidence is necessary, full natural-language feedback yields the strongest reviewer among handwritten prompting, MIPROv2, and scalar/category GEPA variants, and a guarded verifier derived from the learned policy improves further. Later SEC follow-up comments provide external validation for the benchmark labels, suggesting that regulator-style feedback is a useful learning signal for evidence-grounded compliance reviewers.

1 Introduction

Financial NLP has rapidly moved toward LLM systems for long documents, domain-specific reasoning, and decision support. Models and benchmarks such as BloombergGPT, FinGPT, FinBen, FinanceBench, FinQA, TAT-QA, and ConvFinQA show both the promise of financial LLMs and the continuing difficulty of grounding answers in financial evidence (Wu et al., 2023; Yang et al., 2023; Xie et al., 2024; Islam et al., 2023; Chen et al., 2021; Zhu et al., 2021; Chen et al., 2022). Recent FinNLP work has similarly emphasized retrieval-augmented financial QA, SEC filing QA, financial

intelligence benchmarks, factuality judging, and evasive-answer detection (Lithgow-Serrano et al., 2025; Lai et al., 2025; Klimaszewski et al., 2025; Wu et al., 2025; Al Nuaimi et al., 2025). Most such settings, however, still present a static input and ask for a static answer, label, or report.

Many operational finance and compliance workflows have a different structure. A regulator, auditor, lender, or control function requests a correction, explanation, disclosure, or supporting artifact; the organization responds that it has revised an external document; the reviewer must decide whether visible evidence actually satisfies the request. We call this class of problems evidence-grounded response resolution. It is related to fact verification and document-level legal NLI (Thorne et al., 2018; Koreeda and Manning, 2021), but adds a prior reviewer request, a response that may only claim a revision, revised evidence that must be checked, and textual feedback about what remains missing.

Existing work leaves three limitations for this setting. First, benchmarks rarely ask whether an organization has repaired a documented deficiency. Second, response evaluation often has weak obligation grounding: a response can sound persuasive while amended evidence omits a named disclosure, quantification, exhibit, legal basis, or accounting analysis. Third, prompt and pipeline optimization often discards feedback structure by receiving only scalar success, although review failures are naturally described in text: what was requested, what was claimed, what evidence showed, and what gap remained.

We study these limitations through U.S. Securities and Exchange Commission (SEC) comment-letter review. SEC staff issue comments requesting expanded disclosure, clarification, revised accounting discussion, removal of unsupported statements, or additional exhibits; companies respond in writing, often with statements such as “the disclosure has been revised accordingly.” After review

completion, the SEC releases staff comment letters and company responses through EDGAR. These traces often point to amended filings and sometimes include later same-topic SEC follow-up comments. Accounting research treats comment letters as economically meaningful regulatory signals (Cassell et al., 2013; Bozanic et al., 2017; Baugh and Schmardebeck, 2023; Ryans, 2021); for NLP, they provide observable request-response-evidence-feedback traces.

We introduce RegTrace-Agent, a framework for evidence-grounded reviewers. It reconstructs review threads, retrieves revised-document evidence, decomposes requests into obligations, adjudicates visible evidence gaps, compiles scalar/category/full feedback, and optimizes a guarded reviewer policy. The learned reviewer is not a single hand-tuned prompt; it is the policy inside a system that gathers evidence, checks obligations, emits a guarded verdict, and uses adjudicated failures to improve the next offline reviewer.

This design makes response resolution a strong testbed for language-feedback optimization. Prompt-programming work treats LLM pipelines as declarative programs whose prompts and demonstrations can be optimized (Khattab et al., 2023); GEPA argues that natural-language feedback can expose reusable failure structure that scalar reward discards (Agrawal et al., 2026). In our setting, that structure is regulator-style: a mistake can be explained as a missing requested disclosure, unsupported revision claim, absent exhibit, or incomplete analysis.

The above limitations motivate three research questions. RQ1 asks whether public review traces can be converted into a benchmark for evidence-grounded response resolution. RQ2 asks whether natural-language feedback improves reviewer optimization beyond scalar prompt search and coarse categories. RQ3 asks whether the resulting decisions generalize across threads, time, and issue categories, and whether unresolved decisions are corroborated by later regulator follow-up.

We make four contributions. First, we define evidence-grounded response resolution and introduce RegTrace-Agent, a closed-loop framework for evidence retrieval, obligation checking, feedback construction, and reviewer optimization. Second, we instantiate it as RegTrace-SEC, a public EDGAR benchmark with SEC comments, company responses, amended-filing evidence, labels, and feedback fields. Third, we show that

GEPA with full natural-language feedback outperforms handwritten prompting, MIPROv2, and scalar/category GEPA variants; a guarded verifier distilled from the learned policy improves further. Fourth, we provide evidence ablations, feedback-component ablations, prompt-evolution analysis, and systematic corroboration from real next-round SEC follow-up comments.

2 Related Work

Financial NLP and evidence-grounded evaluation. Financial NLP has moved from broad document classification toward long-document reasoning, evidence grounding, and compliance-sensitive decisions. FinQA, TAT-QA, and ConvFinQA introduced financial reasoning questions over reports, tables, and conversations (Chen et al., 2021; Zhu et al., 2021; Chen et al., 2022); FinanceBench evaluates open-book QA over company documents (Islam et al., 2023). Recent FinNLP work has similarly emphasized RAG over financial documents, accessible finance-intelligence benchmarks, factuality and causality judging, SEC filing QA, and evasive-answer detection (Lithgow-Serrano et al., 2025; Klimaszewski et al., 2025; Lai et al., 2025; Al Nuaimi et al., 2025; Wu et al., 2025). Our task differs by asking whether retrieved evidence satisfies a prior regulator request.

SEC comment letters and regulatory review. Accounting and disclosure research has studied SEC comment letters as regulatory signals, including remediation costs, disclosure effects, recurring disclosure deficiencies, and textual classification of comment-letter topics (Cassell et al., 2013; Bozanic et al., 2017; Baugh and Schmardebeck, 2023; Ryans, 2021). That literature generally treats comment letters as economic or disclosure events. We instead treat the correspondence as document-grounded supervision for LLM systems. Each review unit contains a staff request, company response, amended filing evidence, and sometimes a next-round SEC follow-up.

Prompt optimization and language feedback. DSPy reframes LLM pipelines as declarative programs whose prompts and demonstrations can be optimized against metrics (Khattab et al., 2023). MIPROv2 and related optimizers primarily use scalar feedback. A broader line studies language feedback, including self-refinement, verbal reinforcement, and critique-based training (Scheurer

et al., 2022; Madaan et al., 2023; Shinn et al., 2023). GEPA uses natural-language reflection over execution traces (Agrawal et al., 2026). This distinction is central here because SEC response errors are naturally textual: missing disclosures, unsupported revision claims, absent exhibits, incomplete analysis, or over-demand beyond the SEC request.

3 RegTrace-Agent Framework

3.1 General Review-Trace Setting

RegTrace-Agent applies to workflows where a response claims to satisfy a prior review request by modifying or supplying evidence outside the response itself. Formally, each review trace contains a request q , a response r , a collection of revised or supporting artifacts $\mathcal{D}$, and optional later reviewer feedback u^+ . The system retrieves evidence $e = R(q, r, \mathcal{D})$, extracts obligations $o_{1:m} = O(q)$, and applies a verifier $V(q, r, e, o_{1:m})$ whose output is either resolved or unresolved. The same trace can also produce feedback for learning. A scalar signal says whether the prediction is correct. A category signal names a broad issue or gap type. A full natural-language signal describes the request, the response action, the visible evidence, and any unmet requirement. This makes the framework applicable beyond SEC review whenever request-response-evidence traces are available.

3.2 SEC Instantiation

Each SEC example contains three test-time fields: `sec_comment`, `company_response`, and `retrieved_snippets` from amended filings. The output is binary. `resolved` means visible amended evidence covers the material request; `unresolved` means a named item, quantification, exhibit, document, legal analysis, accounting analysis, or other material requirement remains missing. The label is a visible-evidence judgment, not a legal determination or an inference from absence of later follow-up.

3.3 SEC Trace Construction

Figure 1 separates the runtime reviewer from the feedback-learning loop. In the SEC instantiation, we harvest UPLOAD and CORRESP materials from EDGAR, group staff comments and company responses into review threads, and select cases where the company claims a disclosure was added, revised, removed, or filed elsewhere. We identify amended filing candidates using filing metadata,

Quantity	Count
Examples	472
SEC review threads	109
Companies	95
Resolved / unresolved	153 / 319
Examples with verified follow-up	236
Issue categories	7
Visible-evidence gap types	8

Table 1: Benchmark scale. Issue categories include other disclosure issues, risk factors, SPAC disclosure, revenue recognition, crypto, non-GAAP, and cybersecurity.

amendment references, page and section cues, and response text. Candidate filings are drawn from a window of 10 days before to 60 days after the response date, with at most 8 candidate filings per row. A local lexical ranker retrieves top chunks, and an LLM reranker/adjudicator evaluates the top 3 snippets. From 589 revision-claim cases, 588 have candidate filings and snippets; 472 remain after requiring visible evidence that directly or partially addresses the SEC request. We then adjudicate whether visible evidence satisfies the request and construct labels plus feedback fields.

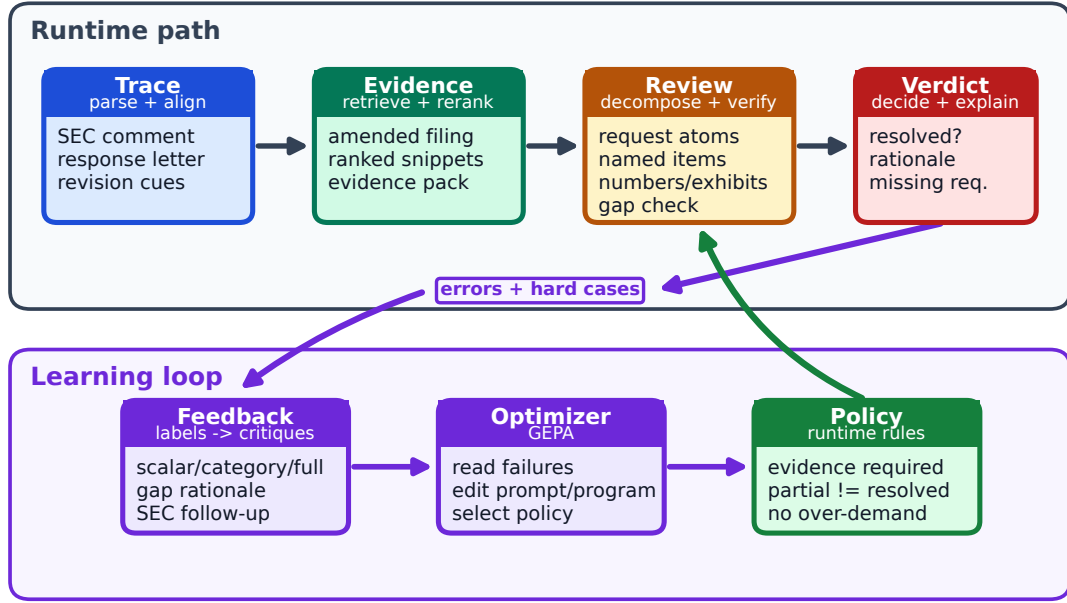
The retrieval component gives the agent evidence to inspect. The adjudication and feedback layer gives it something to learn from: what the reviewer asked for, what the company claimed to do, what the visible evidence supports, and what remains missing. These adjudication fields are not shown as ordinary test-time inputs. They are used for optimization feedback, audits, and analysis.

3.4 Scale, Categories, and Splits

RegTrace-SEC contains 472 examples from 109 SEC review threads and 95 companies. The benchmark focuses on difficult revision-claim cases, and the label distribution is operationally realistic: 153 examples are resolved, and 319 are unresolved. Table 1 summarizes the corpus and split regimes.

The corpus covers seven issue categories and eight visible-evidence gap types; Appendix B gives the full distribution. We use grouped, temporal, and topic-holdout regimes. The grouped split holds out entire review threads (229/104/139 train/dev/test). The temporal split trains on 2022–2023 and tests on 2024–2025 (264/64/144). The topic-holdout split tests transfer to held-out issue categories (279/68/125), including crypto and non-GAAP cases absent from training categories.

RegTrace-Agent: closed-loop evidence review and feedback optimization



Test-time inputs: SEC comment, company response, retrieved amended evidence. Feedback fields are train/audit only.

Figure 1: RegTrace-Agent architecture. The runtime path parses a review trace, retrieves amended-filing evidence, checks request obligations, and emits a guarded verdict. The learning loop stores scalar, category, and natural-language feedback, lets GEPA revise the reviewer policy, and reinjects the learned policy into the obligation-review step. At test time the reviewer sees only the SEC comment, company response, and retrieved amended evidence.

3.5 From Labels to Feedback

A standard benchmark would stop at binary labels. Our construction preserves the textual structure of each decision. For every example, adjudication records the material SEC request, the company action, evidence relevance, supporting evidence or resolution basis, visible unmet requirement, and a short rationale. This lets us compare increasingly rich optimization signals:

- scalar: correct or incorrect;
- category: correct/incorrect plus coarse issue or gap type;
- full: natural-language feedback describing the SEC request, company action, evidence coverage, and unmet requirement.

Full feedback can state that the evidence discusses combined revenue projections but fails to disclose entity-level operating costs, product pricing, gross margins, and forecast limitations. This is the reusable decision rule that scalar metrics discard.

4 Methods

4.1 Prediction Setup

The prediction model receives the SEC comment, company response, and raw retrieved amended-filing snippets. It returns a binary label and a short rationale. Explicit resolved is scored as resolved, unresolved, partially_resolved, and other insufficient labels are normalized to unresolved. The rationale is required for inspection but not directly rewarded.

4.2 Optimization Ladder

We compare five methods. Baseline prompt is a handwritten instruction. MIPROv2 is scalar prompt optimization. GEPA-scalar uses reflective prompt optimization with correctness-only feedback. GEPA-category adds coarse category feedback. GEPA-full uses full natural-language adjudication feedback.

All methods receive the same evidence at test time, isolating the optimization signal rather than evidence access. The key comparison is whether natural-language feedback adds value beyond prompt search itself.

4.3 Implementation Details

All main prompt-optimization runs use openai/gpt-4o-mini as the frozen prediction model with temperature 0 and 500 output tokens. GEPA uses the same model as the reflection model with temperature 0.7 and 1500 reflection tokens, seed 17, max_metric_calls=150, reflection minibatch size 4, Pareto candidate selection, and 4 evaluation threads. MIPROv2 uses the same prediction and prompt models, 8 trials, 4 prompt candidates, and at most 4 labeled and 4 bootstrapped demonstrations. The evidence reranker/adjudicator also uses gpt-4o-mini at temperature 0. Model aliases rather than dated provider snapshots are recorded in the released artifacts.

4.4 Optimization as Feedback-Conditioned Search

We view each method as search over a prompt or prompt program p . Given an input x and label y , the frozen LLM defines $\hat{y}_p(x) = f_\theta(x; p)$, and optimization seeks a prompt with high validation reward. The methods differ in what they observe after an error. Scalar optimization observes only correctness. Category feedback adds a coarse gap type. Full feedback adds a short critique describing the request, company action, visible evidence, and unmet requirement.

The intuition is information-theoretic. Let C be the latent cause of failure, such as missing quantification or unsupported revision claim. Full feedback should help when it lowers uncertainty about C , because the optimizer can propose a more targeted semantic edit. This is not a general guarantee: a noisy or misaligned language channel can hurt. The hypothesis of RegTrace-Agent is that review traces are unusually well suited to language feedback because the failure causes are naturally written as reviewer critiques.

4.5 What the Optimizer Sees

During optimization, each candidate prompt is evaluated on development examples. When a prediction is incorrect, the metric returns feedback in one of three forms:

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scalar: incorrect
category: incorrect;
gap_type=missing_specific_disclosure
full: incorrect. The SEC requested
X. The company claimed Y. The visible
evidence shows Z, but does not show
required item W.
```

Input	Acc.	Ma-F1	Res-F1	Unres-F1
Response only	0.712	0.575	0.333	0.817
+ raw snippets	0.784	0.748	0.651	0.844
+ oracle summary	0.964	0.956	0.938	0.975

Table 2: Evidence ablation. Adding raw amended-filing snippets raises macro-F1 by 17.3 points. The oracle summary is an upper bound, not a deployable baseline.

Full feedback summarizes the adjudicated relationship between the SEC request, response, and retrieved evidence for optimization examples only. It does not reveal hidden evidence at test time; any advantage must come from a better learned checklist for visible-evidence review.

5 Experiments

5.1 Evidence Ablation

We first test whether amended-filing evidence matters. Table 2 evaluates a response-only condition, a raw-evidence condition, and an oracle evidence-summary condition on the grouped held-out test set.

Response-only prompting cannot verify whether a revision claim is supported. Raw snippets nearly double resolved F1, while the oracle summary shows that much of the remaining difficulty is interpreting snippets against a specific request.

5.2 Main Prompt-Optimization Results

Table 3 reports the final prompt-optimization ladder on the frozen grouped split. All methods receive the same test-time context.

The majority baseline has high accuracy because unresolved examples dominate, but zero resolved F1. The handwritten prompt is conservative: it has perfect unresolved precision but misses most gaps. MIPROv2 improves substantially, GEPA-scalar improves recall, and GEPA-category is too coarse in this run. GEPA-full is strongest overall, improving macro-F1 by 10.4 points over MIPROv2 and 8.1 points over GEPA-scalar.

We also compute five-fold out-of-fold results over review-thread groups. Mean macro-F1 is 0.698 for GEPA-full, 0.647 for MIPROv2, 0.609 for GEPA-scalar, 0.545 for GEPA-category, and 0.446 for the handwritten baseline. The fold standard deviations are nontrivial (0.151 for GEPA-full and 0.121 for MIPROv2), so we treat the single grouped split as a headline comparison and the out-of-fold analysis as stability evidence rather than as a deterministic margin claim.

Method	Acc.	Macro-F1	Res-F1	Unres-F1	Unres-P	Unres-R
Majority unresolved	0.691	0.409	0.000	0.817	0.691	1.000
Baseline prompt	0.424	0.402	0.518	0.286	1.000	0.167
MIPROv2	0.655	0.652	0.619	0.684	0.929	0.542
GEPA-scalar	0.705	0.675	0.577	0.773	0.824	0.729
GEPA-category	0.597	0.595	0.562	0.627	0.870	0.490
GEPA-full	0.784	0.756	0.674	0.839	0.867	0.812

Table 3: Frozen grouped-split leaderboard. GEPA-full gives the best macro-F1 and unresolved F1 under the same raw-evidence test-time context.

Reviewer	Macro-F1	Res-F1	Unres-F1
Monolithic reviewer	0.645	0.614	0.676
Obligation-guided reviewer	0.619	0.613	0.624
GEPA-full optimized prompt	0.756	0.674	0.839
RegTrace guarded verifier	0.801	0.727	0.874

Table 4: Reviewer-structure ablation. The guarded verifier is a policy-distillation prototype derived from the obligation-evidence policy exposed by GEPA-full prompt evolution.

5.3 Reviewer-Structure Ablation

The optimization ladder tests feedback richness. We also ask whether reviewer structure matters. Table 4 compares a monolithic judgment, an obligation-guided judgment, and a guarded verifier that requires every material obligation to be visibly satisfied, with GEPA-full as the optimized-prompt reference.

Simply listing obligations is not enough; the gain comes from the guard that requires visible support for each material request. We report this as a policy-distillation prototype rather than the primary optimizer comparison, since it adds human-designed structure after prompt-evolution analysis.

5.4 Feedback Component Ablation

Table 5 holds test-time input fixed and changes only optimization-time feedback.

Request-action-only feedback induces aggressive skepticism and the highest unresolved recall, but sacrifices resolved F1. Full feedback teaches both sides of the boundary: unmet requirements for unresolved cases and supporting evidence for resolved cases.

5.5 Temporal and Topic Generalization

Table 6 shows that the optimized prompt transfers beyond the grouped split. In the temporal setting, GEPA-full trains/tunes on 2022–2023 and tests on 2024–2025 material. In the topic setting, test examples include crypto and non-GAAP cases absent from training issue categories.

The topic-holdout result argues against a curated topic explanation. Per-topic breakdowns show GEPA-full improves over MIPROv2 on crypto, non-GAAP, revenue recognition, risk factors, and broad other-disclosure cases, with largest gains in revenue recognition and non-GAAP.

6 External SEC Follow-Up Corroboration

Visible-evidence labels come from an adjudication pipeline, so we test whether they align with later regulator behavior. Verified same-topic next-round SEC follow-up is not the primary label, but it is a useful corroboration signal because it can restate or narrow an unresolved gap.

We additionally audit 60 examples with an independent LLM auditor that receives the SEC comment, company response, and visible amended evidence but not the original label rationale. The audit agrees with the benchmark label in 49 of 60 cases, with mean confidence 0.897. We report this as an audit of visible-evidence label consistency, not as expert legal/accounting validation.

Among 236 examples with verified follow-up, unresolved cases are corroborated at 68.1%, while resolved cases are corroborated at only 9.2% (Table 7). Follow-up availability itself is balanced across labels (49.7% for resolved and 50.2% for unresolved; Fisher exact $p = 1.0$), so this is not merely an availability artifact.

Following reviewer-style paired analysis, we also treat later follow-up as a noisy external reference on a clean 220-example subset. The benchmark label reaches 0.756 macro-F1 against this signal, compared with 0.590 for five-fold out-of-fold GEPA-full predictions and 0.460 for the handwritten baseline. We therefore use follow-up primarily as label-validity evidence, not as proof that every GEPA-full win is separately corroborated beyond the unresolved base rate.

In one expense-disclosure case, the SEC asked for material components and trend discussion. The

Feedback	Acc.	Macro-F1	Res-F1	Unres-F1	Unres-P	Unres-R
Full feedback	0.784	0.756	0.674	0.839	0.867	0.812
No evidence rationale	0.712	0.682	0.583	0.780	0.826	0.740
No unmet requirement	0.741	0.708	0.609	0.806	0.833	0.781
Request + action only	0.784	0.724	0.595	0.853	0.806	0.906

Table 5: Feedback component ablation. Request-action feedback induces aggressive unresolved detection, while full feedback gives the best macro-F1 by calibrating skepticism with evidence rationales.

Split	Method	Acc.	Macro-F1	Unres-F1	Unres-R
Time	Baseline	0.451	0.434	0.336	0.206
	MIPROv2	0.521	0.519	0.489	0.340
	GEPA-scalar	0.653	0.650	0.679	0.546
	GEPA-full	0.826	0.791	0.877	0.918
Topic	Baseline	0.440	0.440	0.426	0.274
	MIPROv2	0.576	0.565	0.635	0.484
	GEPA-scalar	0.528	0.524	0.569	0.411
	GEPA-full	0.784	0.734	0.849	0.800

Table 6: Temporal and topic-holdout generalization. GEPA-full transfers substantially better than scalar optimization.

Group	N	D/P	Rate
Unres. + f/u	160	109	68.1%
Res. + f/u	76	7	9.2%

Table 7: Real follow-up corroboration. Later same-topic SEC comments directly or partially corroborate unresolved visible-evidence labels far more often than resolved labels.

Method	Claim insuff.	Evidence required	Request match	Missing items
GEPA-full	2/2	2/2	2/2	2/2
GEPA-scalar	2/2	2/2	0/2	0/2
GEPA-category	2/2	2/2	0/2	0/2
Req.+action only	2/2	2/2	2/2	2/2

Table 8: Rule presence in optimized prompt candidates. Full feedback more consistently induces named-request matching and missing-item reasoning.

Scalar and category feedback learn general cau-490
tion: do not trust a revision claim unless evidence 491
supports it. Full feedback more consistently learns 492
named-request matching and missing-item reason-493
ing (Table 8), shifting from generic skepticism to 494
obligation-level comparison: SEC request → com-495
pany action → visible evidence → remaining gap. 496

7.2 Method-Disagreement Cases

The strongest GEPA-full wins involve concrete 498
request-evidence mismatches. In a financial-499
projections case, the SEC requested individual-500
company forecast information and material assump-501
tions; the amended evidence showed only com-502
bined revenue projections. The baseline treated 503
this as sufficient, while GEPA-full marked it unre-504
solved because the requested assumption set was 505
still absent. 506

In a crypto terminology case, the amended evi-507
dence defined stablecoins but did not explain the 508
SEC’s requested contrast with unstable cryptocur-509
rency. Baseline, MIPROv2, GEPA-scalar, and 510
GEPA-category predicted resolved; GEPA-full 511
predicted unresolved. In a PIPE financing case, 512
MIPROv2 treated the presence of price terms as 513
sufficient; GEPA-full required equivalent price-per-514
share disclosure for each agreement, a gap later 515
echoed in SEC follow-up. 516

GEPA-full also has informative errors. In one 517
ownership-percentage case, retrieved evidence con-518
tained calculations under redemption scenarios and 519
the benchmark label was resolved. GEPA-full 520
predicted unresolved because it wanted a more 521

476 company claimed a revision, but retrieved evidence
477 showed only a line item and totals; GEPA-full
478 predicted unresolved, and the next-round SEC
479 follow-up again requested the component breakout.
480 SPAC transaction-background cases show the same
481 pattern for missing negotiation details.

7 Prompt Evolution and Case Studies

7.1 Prompt Evolution

484 Prompt evolution shows that GEPA-full does not
485 merely make the prompt longer. It turns adjudica-
486 tion feedback into a compact regulatory reading
487 policy. The seed prompt is a single sentence; the
488 final GEPA-full run saves optimized candidates of
489 349 and 397 words (Appendix A).

explicit cover-page presentation. This reveals the main calibration risk: full-feedback prompts learn to be skeptical of unsupported revision claims, but can over-require exact presentation when visible evidence substantially satisfies the request. Appendix C lists additional analyzed cases.

8 Discussion

The main result is not simply that one optimizer beats another. RegTrace-Agent turns a recurring compliance pattern into an offline feedback-learning system: collect the trace, retrieve the revised artifact, check obligations against visible evidence, issue a guarded verdict, and recycle adjudicated failures as feedback. The output is not only a binary label, but a traceable unresolved-gap rationale.

Language feedback helps because it preserves decision structure. A scalar signal says only that a verdict failed; full feedback says which named request, company action, amended snippet, and remaining obligation were mismatched. The optimized prompt is consequently an interpretable policy artifact: extract material requests, distrust unsupported revision claims, inspect evidence for named items, avoid over-requiring beyond the regulator’s request, and distinguish partial disclosure from resolution.

SEC correspondence is the public setting where we can observe request, response, revised filing, and later reviewer behavior, but the trace structure is broader. Audit remediation, regulatory reporting, lending-document stipulations, data-quality exception handling, and internal compliance monitoring also contain request-response-evidence loops. The transferable claim is that such traces can support retrieval, obligation checking, adjudication feedback, optimized reviewing, and hard-case mining; the limitation is that high-recall language-feedback policies can become over-skeptical.

9 Conclusion

We introduced RegTrace-Agent, a framework for evidence-grounded regulatory response review, and instantiated it as RegTrace-SEC using public SEC comment-letter traces. Across grouped, temporal, and topic-holdout evaluations, full natural-language feedback yields the strongest optimized reviewer. Evidence ablations, prompt evolution, and real SEC follow-up corroboration suggest that financial NLP benchmarks can move from static

labels toward feedback-rich review systems that preserve the reasons behind expert decisions.

Limitations and Data Statement

This study is conditioned on visible retrieved evidence, so missed snippets can affect labels. The benchmark is LLM-adjudicated, with independent audit and external follow-up corroboration but without a full expert legal/accounting audit. Same-topic SEC follow-up is a noisy behavioral signal, not ground truth. We frame labels as visible-evidence judgments rather than legal determinations. The release includes benchmark JSONL, splits, accession identifiers, retrieval metadata, snippets, labels, feedback fields, prompt versions, outputs, and analysis scripts.

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722	A Full Optimized Prompts	
723	The main paper quotes short excerpts from opti-	
724	mized prompts. Below are the full GEPA-full op-	
725	timized candidates exposed by the final grouped-	
726	split run. The seed prompt was: “Classify whether	
727	a company resolved an SEC comment using the	
728	visible evidence.”	
729	A.1 GEPA-full Optimized Candidate 1	
730	You are tasked with classifying whether a com-	
731	pany has resolved a comment from the SEC based	
732	on the provided evidence. Follow these instruc-	
733	tions carefully:	
734	1. Input Format: You will receive three types of	
735	inputs: <code>sec_comment</code> , <code>company_response</code> , and	
736	<code>amended_evidence</code> .	
737	2. Classification Criteria: Determine if the com-	
738	pany has fully resolved the SEC comment by ana-	
739	lyzing the company response in conjunction with	
740	the amended evidence. If the response satisfies	
741	the SEC’s request and the evidence supports this	
742	claim, classify the comment as resolved. If the re-	
743	sponse does not meet the SEC’s request or if there	
744	are still outstanding issues, classify the comment	
745	as unresolved.	
746	3. Evaluation Process: Carefully compare the	
747	SEC request against the company’s response and	
748	the amended evidence. Look for specific items or	
749	information requested by the SEC, quantification	
750	or detailed analysis that meets SEC requirements,	
751	references to regulatory guidance or previous de-	
752	cisions, and clear explanations of how the com-	
753	pany adjusted disclosures or practices. Take note	
754	of visible unmet requirements, partial fixes, or	
755	lack of supporting documentation in the amended	
756	evidence.	
	4. Output Format: Include a reasoning section,	757
	a label indicating resolved or unresolved, and a	758
	concise reason.	759
	5. Feedback Incorporation: Be vigilant about	760
	claims made by the company; ensure that	761
	amended evidence corroborates these claims.	762
	Identify instances where the response may par-	763
	tially address SEC concerns but still falls short of	764
	complete compliance.	765
	A.2 GEPA-full Optimized Candidate 2	766
	You are tasked with classifying whether a com-	767
	pany has resolved a comment issued by the SEC	768
	based on provided evidence. Follow these specific	769
	instructions for the task:	770
	1. Input Format: You will receive <code>sec_comment</code> ,	771
	<code>company_response</code> , and <code>amended_evidence</code> .	772
	The SEC comment is the original request for clar-	773
	ification or additional information. The company	774
	response indicates what changes were made or	775
	explanations provided. The amended evidence	776
	contains snippets from the amended filing or other	777
	documents that support the response.	778
	2. Classification Criteria: Determine whether the	779
	company’s response and amended evidence ade-	780
	quately address the SEC’s concerns as outlined	781
	in the original comment. A response is resolved	782
	if it fully satisfies the requirements of the SEC	783
	comment, including clear explanations, specific	784
	details, and quantifications when necessary. A	785
	response is unresolved if there are still gaps or	786
	missing elements in the company’s explanation or	787
	if amended evidence does not address all parts of	788
	the SEC comment.	789
	3. Evaluation Process: Carefully analyze the SEC	790
	comment and identify the specific details or expla-	791
	nations requested. Review the company response	792
	to see if the company acknowledged the comment	793
	and provided a meaningful response. Examine	794
	amended-evidence snippets to confirm that they	795
	support the company’s claims and cover all re-	796
	quested aspects of the SEC comment. Look for	797
	direct references in the amended evidence that	798
	address the SEC’s specific requests, such as own-	799
	ership structures, accounting adjustments, or en-	800
	forcement risks.	801
	4. Labeling: If the company has fully addressed	802
	the SEC’s concerns with adequate evidence, label	803
	the response resolved. If there are omissions or	804
	inadequacies in addressing SEC requests, label it	805
	unresolved.	806
	5. Justification: Provide clear and concise rea-	807
	soning that cites specific elements from the SEC	808
	comment, company response, and amended evi-	809
	dence. Highlight specific unmet requirements or	810
	gaps in the evidence that contribute to an unre-	811
	solved classification.	812
	6. Feedback Incorporation: Pay attention to feed-	813
	back from previous examples to identify common	814
	pitfalls, such as failing to compare the SEC re-	815
	quest against the evidence adequately or missing	816
	key elements in the response.	817
	B Additional Dataset Details	818
	The most common issue categories are other disclo-	819
	sure issues (183), risk factors (91), SPAC disclosure	820

821 (78), revenue recognition (75), crypto (28), non-
822 GAAP (16), and cybersecurity (1). The most com-
823 mon gap types are resolved visible evidence (153),
824 missing specific disclosure (140), missing quantifi-
825 cation (59), resolved-but-regulator-requested-more-
826 detail (38), missing exhibit or document (29), in-
827 complete accounting analysis (25), incomplete le-
828 gal or regulatory analysis (25), and other visible
829 evidence gap (3).

830 **C Additional Qualitative Cases**

831 The qualitative appendix used during analysis in-
832 cludes full SEC comments, company responses,
833 amended evidence snippets, method predictions,
834 GEPA-full rationales, and later SEC follow-up text
835 for selected cases. These include stablecoin termi-
836 nology, financial forecast assumptions, PIPE price-
837 per-share disclosure, Item 404 related-party disclo-
838 sure, expense breakout and trend discussion, and
839 SPAC transaction-background disclosure.